

For the use of a Registered Medical Practitioner only

PRESCRIBING INFORMATION

PROART Tablets 62.5 mg and PROART Tablets 125 mg (Bosentan Monohydrate Tablets)

COMPOSITION

PROART 62.5 mg

Each film coated tablet contains:

Bosentan Monohydrate equivalent to

Bosentan 62.5 mg

PROART 125 mg

Each film coated tablet contains:

Bosentan Monohydrate equivalent to

Bosentan 125 mg

Excipients: Maize starch, Pregelatinized starch, Sodium starch glycolate, Povidone, Purified water, Glyceryl behenate, Magnesium stearate, Isopropyl alcohol and Opadry 21K520019 (Yellow) containing Hypromellose, Titanium Dioxide, Triacetin, Talc, Ethylcellulose, Iron Oxide Yellow, Hpmc /Hypromellose and Iron Oxide Red.

DESCRIPTION

For 62.5 mg

Light peach to peach colored film coated round shaped biconvex tablet debossed with '62.5' on one side and plain the other side.

For 125 mg

Light peach to peach colored film coated Oval shaped biconvex tablet debossed with '125' on one side and plain the other side.

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

- Pharmacodynamics

Pharmacotherapeutic group: other antihypertensives, ATC code: C02KX01

Mechanism of action

Bosentan is a dual endothelin receptor antagonist (ERA) with affinity for both endothelin A and B (ET_A and ET_B) receptors. Bosentan decreases both pulmonary and systemic vascular resistance resulting in increased cardiac output without increasing heart rate.

The neurohormone endothelin-1 (ET-1) is one of the most potent vasoconstrictors known and can also promote fibrosis, cell proliferation, cardiac hypertrophy and remodelling, and is pro-inflammatory. These effects are mediated by endothelin binding to ET_A and ET_B receptors located in the endothelium and vascular smooth muscle cells. ET-1 concentrations in tissues and plasma are increased in several cardiovascular disorders and connective tissue diseases, including PAH, scleroderma, acute and chronic heart failure, myocardial ischaemia, systemic hypertension and atherosclerosis, suggesting a pathogenic role of ET-1 in these diseases. In PAH and heart failure, in the absence of endothelin receptor antagonism, elevated ET-1 concentrations are strongly correlated with the severity and prognosis of these diseases.

Bosentan competes with the binding of ET-1 and other ET peptides to both ET_A and ET_B receptors, with a slightly higher affinity for ET_A receptors ($K_i = 4.1\text{--}43$ nanomolar) than for ET_B receptors ($K_i = 38\text{--}730$ nanomolar). Bosentan specifically antagonises ET receptors and does not bind to other receptors.

- **Pharmacokinetics**

Bosentan is reported to have dose- and time-dependent pharmacokinetics. Clearance and volume of distribution is reported to decrease with increased intravenous doses and increase with time. After oral administration, the systemic exposure is reported to be proportional to dose up to 500 mg. At higher oral doses, $C_{\max}$ and AUC is reported to increase less than proportionally to the dose.

Absorption

The absolute bioavailability of bosentan is approximately 50% which is not affected by food. The maximum plasma concentrations are attained within 3–5 hours.

Distribution

Bosentan is highly bound (>98%) to plasma proteins, mainly albumin. The volume of distribution (V_{ss}) is about 18 litres. Bosentan does not penetrate into erythrocytes.

Biotransformation and elimination

The clearance is 8.2 L/h after a single intravenous dose of 250 mg. The terminal elimination half-life ($t_{1/2}$) is 5.4 hours.

Upon multiple dosing, plasma concentrations of bosentan decrease gradually to 50–65% of those reported after single dose administration, probably due to auto-induction of metabolising liver enzymes. Steady-state conditions are reached within 3–5 days.

Bosentan is eliminated by biliary excretion following metabolism in the liver by the cytochrome P450 isoenzymes, CYP2C9 and CYP3A4. Less than 3% of an administered oral dose is recovered in urine.

Bosentan forms three metabolites and only one of these is pharmacologically active. This metabolite is mainly excreted unchanged via the bile. In patients with evidence of the presence of cholestasis, the exposure to the active metabolite may be increased.

Bosentan is an inducer of CYP2C9 and CYP3A4 and possibly also of CYP2C19 and the P-glycoprotein. *In vitro*, bosentan inhibits the bile salt export pump in hepatocyte cultures.

Bosentan has no relevant inhibitory effect on the CYP isoenzymes (CYP1A2, 2A6, 2B6, 2C8, 2C9, 2D6, 2E1, 3A4). Consequently, bosentan is not expected to increase the plasma concentrations of medicinal products metabolised by these isoenzymes.

Pharmacokinetics in special populations

As reported, it is not expected that the pharmacokinetics of bosentan will be influenced by gender, body weight, race, or age in the adult population to any relevant extent.

Children

Due to limited data in children below 2 years of age, pharmacokinetics remains not well characterised in this age category.

Hepatic impairment

In patients with mildly impaired liver function (Child-Pugh class A) no relevant changes in the pharmacokinetics have been reported.

Bosentan is contraindicated in patients with moderate to severe hepatic impairment, i.e., Child-Pugh class B or C (see CONTRAINDICATIONS).

Renal impairment

No dose adjustment is required in patients with renal impairment. There is no specific clinical experience in patients undergoing dialysis. Based on physicochemical properties and the high degree of protein binding, bosentan is not expected to be removed from the circulation by dialysis to any significant extent (see DOSE AND METHOD OF ADMINISTRATION).

INDICATIONS

Treatment of pulmonary arterial hypertension (PAH) in patients of WHO functional class II-IV.

Considerations for use:

Patients with WHO Class II symptoms showed reduction in the rate of clinical deterioration and a trend for improvement in walk distance. Physicians should consider whether these benefits are sufficient to offset the risk of liver injury in WHO Class II patients, which may preclude future use as their disease progresses.

DOSE AND METHOD OF ADMINISTRATION

General

Bosentan treatment should be initiated at a dose of 62.5 mg twice daily for 4 weeks and then increased to the maintenance dose of 125 mg twice daily. Doses above 125 mg twice daily did not appear to confer additional benefit sufficient to offset the increased risk of liver injury.

PROART Tablets should be administered in the morning and evening with or without food.

Management in case of clinical deterioration of PAH

In the case of clinical deterioration (e.g., decrease in 6-minute walk test distance by at least 10% compared with pre-treatment measurement) despite bosentan treatment for at least 8 weeks (target dose for at least 4 weeks), alternative therapies should be considered. However, some patients who show no response after 8 weeks of treatment with bosentan may respond favourably after an additional 4 to 8 weeks of treatment.

In the case of late clinical deterioration despite treatment with bosentan (i.e., after several months of treatment), the treatment should be re-assessed. Some patients not responding well to 125 mg twice daily of bosentan may slightly improve their exercise capacity when the dose is increased to 250 mg twice daily. A careful benefit/risk assessment should be made, taking into consideration that the liver toxicity is dose dependent.

Required Monitoring

Liver aminotransferase levels must be measured prior to initiation of treatment and then monthly. If elevated aminotransferase levels are seen, changes in monitoring and treatment should be initiated.

Dose Adjustment and Monitoring in Patients Developing Aminotransferase Abnormalities

The table below summarizes the dosage adjustment and monitoring recommendations for patients who develop aminotransferase elevations $> 3 \times \text{ULN}$ during therapy with bosentan. If liver aminotransferase elevations are accompanied by clinical symptoms of liver injury (such as nausea, vomiting, fever, abdominal pain, jaundice, or unusual lethargy or fatigue) or increases in bilirubin $\geq 2 \times \text{ULN}$, treatment with bosentan should be stopped. There is no experience with the reintroduction of bosentan in these circumstances.

<i>ALT/AST levels</i>	<i>Treatment and monitoring recommendations</i>
$> 3 \text{ and } \leq 5 \times \text{ULN}$	Confirm by another aminotransferase test; if confirmed, reduce the daily dose or interrupt treatment, and monitor aminotransferase levels at least every 2 weeks. If the aminotransferase levels return to pretreatment values, continue or reintroduce the treatment as appropriate (see below).
$> 5 \text{ and } \leq 8 \times \text{ULN}$	Confirm by another aminotransferase test; if confirmed, stop treatment and monitor aminotransferase levels at least every 2 weeks. Once the aminotransferase levels return to pretreatment values, consider reintroduction of the treatment (see below).
$> 8 \times \text{ULN}$	Treatment should be stopped and reintroduction of bosentan should not be considered. There is no experience with reintroduction of bosentan in these circumstances.

If bosentan is re-introduced it should be at the starting dose; aminotransferase levels should be checked within 3 days and thereafter according to the recommendations above.

Use in Women of Child-bearing Potential

Initiate treatment in females of childbearing potential only after a negative pregnancy test and only in females who are using two reliable methods of contraception. Females who have had a tubal sterilization or a Copper T 380A IUD or LNG 20 IUS inserted do not require other forms of contraception. Effective contraception must be practiced throughout treatment and for one month after stopping bosentan. Females should seek contraceptive advice as needed from a gynecologist or similar expert. Urine or serum pregnancy tests should be obtained monthly in females of childbearing potential taking bosentan.

Dose Adjustment in Renally Impaired Patients

The effect of renal impairment on the pharmacokinetics of bosentan is small and does not require dosing adjustment.

Dose Adjustment in Geriatric Patients

Clinical studies of bosentan did not include sufficient numbers of subjects aged 65 and older to determine whether they respond differently from younger subjects.

Dose Adjustment in Hepatically Impaired Patients

Because there is *in vitro* and *in vivo* evidence that the main route of excretion of bosentan is biliary, liver impairment could be expected to increase exposure (C_{max} and AUC) of bosentan.

Mild liver impairment was shown not to impact the pharmacokinetics of bosentan. The influence of moderate or severe liver impairment on the pharmacokinetics of bosentan has not been evaluated. There are no specific reported data to guide dosing in hepatically impaired patients (see WARNINGS AND PRECAUTIONS), caution should be exercised in patients with mildly impaired liver function. Bosentan should generally be avoided in patients with moderate or severe liver impairment.

Dose Adjustment in Children

Safety and efficacy in paediatric patients have not been established.

Dose Adjustment in Patients with Low Body Weight

In patients with a body weight below 40 kg but who are over 12 years of age, the recommended initial and maintenance dose is 62.5 mg twice daily. There is limited information about the safety and efficacy of bosentan in children between the ages of 12 and 18 years.

Discontinuation of Treatment

There is limited experience with abrupt discontinuation of bosentan. No evidence for acute rebound has been observed. Nevertheless, to avoid the potential for clinical deterioration, gradual dose reduction (62.5 mg twice daily for 3 to 7 days) should be considered.

CONTRAINDICATIONS

- Hypersensitivity to the active substance or to any of the excipients of this product.
- Moderate to severe hepatic impairment, i.e., Child-Pugh class B or C (see Pharmacokinetics)
- Baseline values of liver aminotransferases, i.e., aspartate aminotransferase (AST) and/or alanine aminotransferase (ALT), greater than 3 × the upper limit of normal (ULN; see WARNINGS AND PRECAUTIONS)
- Concomitant use of cyclosporine A (see DRUG INTERACTIONS)
- Pregnancy (see WARNINGS AND PRECAUTIONS and PREGNANCY AND LACTATION)
- Women of childbearing potential who are not using reliable methods of contraception (see WARNINGS AND PRECAUTIONS, DRUG INTERACTIONS and PREGNANCY AND LACTATION)

WARNINGS AND PRECAUTIONS

The efficacy of bosentan has not been reported in patients with severe PAH. Transfer to a therapy that is recommended at the severe stage of the disease (e.g., epoprostenol) should be considered if the clinical condition deteriorates (see DOSE AND METHOD OF ADMINISTRATION).

The benefit/risk balance of bosentan has not been reported in patients with WHO class I functional status of PAH.

Bosentan should only be initiated if the systemic systolic blood pressure is higher than 85 mmHg.

Bosentan has not been reported to have a beneficial effect on the healing of existing digital ulcers.

Liver function

Elevations in liver aminotransferases, i.e., aspartate and alanine aminotransferases (AST and/or ALT), associated with bosentan are dose dependent. Liver enzyme changes typically occur within the first 26 weeks of treatment but may also occur late in treatment (see SIDE EFFECTS/UNDESIRABLE EFFECTS). These increases may be partly due to competitive inhibition of the elimination of bile salts from hepatocytes but other mechanisms, which have not been clearly established, are probably also involved in the occurrence of liver dysfunction. The accumulation of bosentan in hepatocytes leading to cytolysis with potentially severe damage of the liver, or an immunological mechanism, are not excluded. Liver dysfunction risk may also be increased when medicinal products that are inhibitors of the bile salt export pump, e.g., rifampicin, glibenclamide and cyclosporine A (see CONTRAINDICATIONS and DRUG INTERACTIONS), are co-administered with bosentan, but limited reported information are available.

Liver aminotransferase levels must be measured prior to initiation of treatment and subsequently at monthly intervals for the duration of treatment with bosentan. In addition, liver aminotransferase levels must be measured 2 weeks after any dose increase.

Recommendations in the event of ALT/AST elevations

ALT/AST levels	Treatment and monitoring recommendations
> 3 and $\leq 5 \times$ ULN	The result should be confirmed by a second liver test; if confirmed, a decision should be made on an individual basis to continue bosentan, possibly at a reduced dose, or to stop bosentan administration (see DOSE AND METHOD OF ADMINISTRATION). Monitoring of aminotransferase levels should be continued at least every 2 weeks. If the aminotransferase levels return to pre-treatment values continuing or re-introducing bosentan according to the conditions described below should be considered.

> 5 and $\leq 8 \times \text{ULN}$	The result should be confirmed by a second liver test; if confirmed, treatment should be stopped and aminotransferase levels monitored at least every 2 weeks. If the aminotransferase levels return to pre-treatment values re-introducing bosentan according to the conditions described below should be considered.
> $8 \times \text{ULN}$	Treatment must be stopped and re-introduction of bosentan is not to be considered.
In the event of associated clinical symptoms of liver injury, i.e., nausea, vomiting, fever, abdominal pain, jaundice, unusual lethargy or fatigue, flu-like syndrome (arthralgia, myalgia, fever), treatment must be stopped and re-introduction of bosentan is not to be considered. <u>Re-introduction of treatment</u> Re-introduction of treatment with bosentan should only be considered if the potential benefits of treatment with bosentan outweigh the potential risks and when liver aminotransferase levels are within pre-treatment values. The advice of a hepatologist is recommended. Re-introduction must follow the guidelines detailed in DOSE AND METHOD OF ADMINISTRATION. Aminotransferase levels must then be checked within 3 days after re-introduction, then again after a further 2 weeks, and thereafter according to the recommendations above.	

ULN = upper limit of normal

Haemoglobin concentration

Treatment with bosentan has been reported to be associated with dose-related decreases in haemoglobin concentration (see SIDE EFFECTS/UNDESIRABLE EFFECTS). It is recommended that haemoglobin concentrations be checked prior to initiation of treatment, every month during the first 4 months, and quarterly thereafter. If a clinically relevant decrease in haemoglobin concentration occurs, further evaluation and investigation should be undertaken to determine the cause and need for specific treatment. Cases of anaemia requiring red blood cell transfusion have been reported (see SIDE EFFECTS/UNDESIRABLE EFFECTS).

Women of childbearing potential

As bosentan may render hormonal contraceptives ineffective, and taking into account the risk that pulmonary hypertension deteriorates with pregnancy as well as the teratogenic effects reported in animals:

- Bosentan treatment must not be initiated in women of childbearing potential unless they practise reliable contraception and the result of the pre-treatment pregnancy test is negative
- Hormonal contraceptives cannot be the sole method of contraception during treatment with bosentan
- Monthly pregnancy tests are recommended during treatment to allow early detection of pregnancy

For further information see DRUG INTERACTIONS and PREGNANCY AND LACTATION.

Pulmonary veno-occlusive disease

Cases of pulmonary oedema have been reported with vasodilators (mainly prostacyclins) when used in patients with pulmonary veno-occlusive disease. Consequently, should signs of pulmonary oedema occur when bosentan is administered in patients with PAH, the possibility of associated veno-occlusive disease should be considered. Rare reports of pulmonary oedema have been reported in patients treated with bosentan who had a suspected diagnosis of pulmonary veno-occlusive disease.

Pulmonary arterial hypertension patients with concomitant left ventricular failure

It is recommended that patients be monitored for signs of fluid retention (e.g., weight gain), especially if they concomitantly suffer from severe systolic dysfunction. Should this occur, starting treatment with diuretics is recommended, or the dose of existing diuretics should be increased. Treatment with diuretics should be considered in patients with evidence of fluid retention before the start of treatment with bosentan.

Pulmonary arterial hypertension associated with HIV infection

When treatment with bosentan is initiated in patients who require ritonavir-boosted protease inhibitors, the patient's tolerability of bosentan should be closely monitored with special attention, at the beginning of the initiation phase, to the risk of hypotension and to liver function tests. An increased long-term risk of hepatic toxicity and haematological adverse events cannot be excluded when bosentan is used in combination with antiretroviral medicinal products. Due to the potential for interactions related to the inducing effect of bosentan on CYP450 (see DRUG INTERACTIONS), which could affect the efficacy of antiretroviral therapy; these patients should also be monitored carefully regarding their HIV infection.

Pulmonary hypertension secondary to chronic obstructive pulmonary disease (COPD)

An increase in minute ventilation and a decrease in oxygen saturation have been reported in patients with pulmonary hypertension secondary to severe COPD (stage III of GOLD classification) taking bosentan. The most frequent adverse event reported was dyspnoea, which resolved with discontinuation of bosentan.

Concomitant use with other medicinal products

Concomitant use of bosentan and cyclosporine A is contraindicated (see CONTRAINDICATIONS and DRUG INTERACTIONS).

Concomitant use of bosentan with glibenclamide, fluconazole and rifampicin is not recommended. For further details please refer to DRUG INTERACTIONS.

Concomitant administration of both a CYP3A4 inhibitor and a CYP2C9 inhibitor with bosentan should be avoided (see DRUG INTERACTIONS).

Effects on ability to drive and use machines

No specific studies have been reported to assess the direct effect of bosentan on the ability to drive and use machines. However, bosentan may induce hypotension, with symptoms of dizziness or syncope that could affect the ability to drive or use machines.

DRUG INTERACTIONS

Bosentan is an inducer of the cytochrome P450 (CYP) isoenzymes CYP2C9, CYP3A4 and CYP2C19. Consequently, plasma concentrations of substances metabolised by these isoenzymes will be decreased when bosentan is co-administered. The possibility of altered efficacy of medicinal products metabolised by these isoenzymes should be considered. The dosage of these products may need to be adjusted after initiation, dose change or discontinuation of concomitant bosentan treatment.

Bosentan is metabolised by CYP2C9 and CYP3A4. Inhibition of these isoenzymes may increase the plasma concentration of bosentan (see ketoconazole). The influence of CYP2C9 inhibitors on bosentan concentration has not been studied. The combination should be used with caution.

Fluconazole and other inhibitors of both CYP2C9 and CYP3A4

Co-administration with fluconazole, which inhibits mainly CYP2C9, but to some extent also CYP3A4, could lead to large increases in plasma concentrations of bosentan. The combination is not recommended. For the same reason, concomitant administration of both a potent CYP3A4 inhibitor (such as ketoconazole, itraconazole or ritonavir) and a CYP2C9 inhibitor (such as voriconazole) with bosentan is not recommended.

Cyclosporine A

Co-administration of bosentan and cyclosporine A (a calcineurin inhibitor) is contraindicated (see CONTRAINDICATIONS).

Tacrolimus, sirolimus

Co-administration of tacrolimus or sirolimus and bosentan has not been reported in man but co-administration of tacrolimus or sirolimus and bosentan may result in increased plasma concentrations of bosentan in analogy to co-administration with cyclosporine A. Concomitant bosentan may reduce the plasma concentrations of tacrolimus and sirolimus. Therefore, concomitant use of bosentan and tacrolimus or sirolimus is not advisable. Patients in need of the combination should be closely monitored for adverse events related to bosentan and for tacrolimus and sirolimus blood concentrations.

Glibenclamide

Co-administration of bosentan with glibenclamide has been reported to decrease the plasma concentration of glibenclamide (a CYP3A4 substrate), with potential significant decrease of the hypoglycemic effect. In addition, an increased incidence of elevated aminotransferases was also reported in patients receiving concomitant therapy. Both glibenclamide and bosentan inhibit the bile salt export pump, which could explain the elevated aminotransferases. This combination should not be used. No drug-drug interaction data are available with the other sulfonylureas.

Rifampicin

Co-administration of bosentan with rifampicin, a potent inducer of CYP2C9 and CYP3A4, has been reported to decrease the plasma concentrations of bosentan. A significantly reduced effect of bosentan is expected when it is co-administered with rifampicin. Concomitant use of rifampicin and bosentan is not recommended. Data on other CYP3A4 inducers, e.g., carbamazepine, phenobarbital, phenytoin and St. John's wort are lacking, but their concomitant administration is expected to lead to reduced systemic exposure to bosentan. A clinically significant reduction of efficacy cannot be excluded.

Lopinavir + ritonavir (and other ritonavir-boosted protease inhibitors)

When administered concomitantly with lopinavir + ritonavir, or other ritonavir-boosted protease inhibitors, the patient's tolerability of bosentan should be monitored. Appropriate monitoring of the HIV therapy is recommended. Similar effects would be expected with other ritonavir-boosted protease inhibitors.

Other antiretroviral agents

No specific recommendation can be made with regard to other available antiretroviral agents due to the lack of data. Due to the marked hepatotoxicity of nevirapine, which could add to bosentan liver toxicity, this combination is not recommended.

Hormonal contraceptives

A decrease in the AUC of norethisterone 1 mg + ethinyl estradiol 35 mcg has been reported with co-administration of bosentan. Hormone-based contraceptives alone, regardless of the route of administration (i.e., oral, injectable, transdermal or implantable forms), are not considered as reliable methods of contraception (see WARNINGS AND PRECAUTIONS and PREGNANCY AND LACTATION).

Warfarin

Co-administration of bosentan has been reported to decrease the plasma concentrations of both S-warfarin (a CYP2C9 substrate) and R-warfarin (a CYP3A4 substrate). No dose adjustment is needed for warfarin and similar oral anticoagulant agents when bosentan is initiated, but intensified monitoring of INR is recommended, especially during bosentan initiation and the up-titration period.

Simvastatin

Decrease in the plasma concentrations of simvastatin (a CYP3A4 substrate) and its active β -hydroxy acid metabolite has been reported when co-administered with bosentan. The plasma concentrations of bosentan were not reported to be affected by concomitant simvastatin. Monitoring of cholesterol levels and subsequent dosage adjustment should be considered.

Ketoconazole

Co-administration of bosentan with ketoconazole, a potent CYP3A4 inhibitor, has been reported to increase the plasma concentrations of bosentan. No dose adjustment of bosentan is considered necessary. Similar increases in bosentan plasma concentrations are expected with the other potent CYP3A4 inhibitors (such as itraconazole or ritonavir). However, when combined with a CYP3A4 inhibitor, patients who are poor metabolisers of CYP2C9 are at risk of increases in bosentan plasma concentrations that may be of higher magnitude, thus leading to potential harmful adverse events.

Epoprostenol

The $C_{\max}$ and AUC values of bosentan was reported to be similar in patients with or without continuous infusion of epoprostenol after both single- and multiple-dose administration of combination of bosentan and epoprostenol (see Pharmacodynamics).

Sildenafil

A decrease in the sildenafil AUC and an increase in the bosentan AUC were reported when bosentan was co-administered with sildenafil. Caution is recommended in the case of co-administration.

Tadalafil

Bosentan was reported to reduce tadalafil systemic exposure following multiple dose co-administrations. Tadalafil did not affect the exposure (AUC and $C_{\max}$) of bosentan or its metabolites.

Digoxin

Co-administration of bosentan with digoxin was reported to decrease the AUC, $C_{\max}$ and $C_{\min}$ of digoxin. This interaction is unlikely to be of clinical relevance.

Paediatric population

Interaction studies have only been reported in adults.

SIDE EFFECTS/UNDESIRABLE EFFECTS

The most frequent adverse reactions are headache, oedema/fluid retention, abnormal liver function test and anaemia/haemoglobin decrease. Treatment with bosentan has been associated with dose-dependent elevations in liver aminotransferases and decreases in haemoglobin concentration.

Adverse reactions reported with bosentan are ranked according to frequency. Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness. No clinically relevant differences in adverse reactions were reported.

System organ class	Frequency	Adverse reaction
Blood and lymphatic system disorders	Common	Anaemia, haemoglobin decrease
	Not known	Anaemia or haemoglobin decreases requiring red blood cell transfusion
	Uncommon	Thrombocytopenia
	Uncommon	Neutropenia, leukopenia
Immune system disorders	Common	Hypersensitivity reactions (including dermatitis, pruritus and rash)
	Rare	Anaphylaxis and/or angioedema
Nervous system disorders	Very common	Headache
	Common	Syncope ¹
Eye disorders	Not known	Blurred vision
Cardiac disorders	Common	Palpitations ¹
Vascular disorders	Common	Flushing
	Common	Hypotension ¹
Respiratory, thoracic and mediastinal disorders	Common	Nasal congestion
Gastrointestinal disorders	Common	Gastrooesophageal reflux disease Diarrhoea
Hepatobiliary disorders	Very common	Abnormal liver function test
	Uncommon	Aminotransferase elevations associated with hepatitis (including possible exacerbation of underlying hepatitis) and/or jaundice ²
	Rare	Liver cirrhosis, liver failure ²
Skin and subcutaneous disorders	Common	Erythema
General disorders and administration site conditions	Very common	Oedema, fluid retention

¹ These types of reactions can also be related to the underlying disease.

² Rare cases of unexplained hepatic cirrhosis were reported after prolonged therapy with bosentan in patients with multiple co-morbidities and therapies with medicinal products. There have also been rare reports of liver failure. These cases reinforce the importance of strict adherence to the monthly schedule for monitoring of liver function for the duration of treatment with bosentan

PREGNANCY AND LACTAION

- Pregnancy

Reproductive toxicity has been reported in animals. There are no reliable reported information on the use of bosentan in pregnant women. The potential risk for humans is still unknown. Bosentan is contraindicated in pregnancy (see CONTRAINDICATIONS).

Women of childbearing potential.

Before the initiation of bosentan treatment in women of childbearing potential, the absence of pregnancy should be checked, appropriate advice on reliable methods of contraception provided, and reliable contraception initiated. Patients and prescribers must be aware that due to potential pharmacokinetic interactions, bosentan may render hormonal contraceptives ineffective (see DRUG INTERACTION). Therefore, women of childbearing potential must not use hormonal contraceptives (including oral, injectable, transdermal or implantable forms) as the sole method of contraception but must use an additional or an alternative reliable method of contraception. If there is any doubt about what contraceptive advice should be given to the individual patient, consultation with a gynaecologist is recommended. Because of possible hormonal contraception failure during bosentan treatment, and also bearing in mind the risk that pulmonary hypertension severely deteriorates with pregnancy, monthly pregnancy tests during treatment with bosentan are recommended to allow early detection of pregnancy.

- Lactation

It is not known whether bosentan is excreted into human milk. Breast-feeding is not recommended during treatment with bosentan.

OVERDOSE

Bosentan has been administered as a single dose of up to 2400 mg to healthy subjects and up to 2000 mg/day for 2 months in patients, without any major clinical consequences. The most common adverse reaction was headache of mild to moderate intensity. It has been reported that when bosentan (500 and 1000 mg twice daily) was given concomitantly with cyclosporine A, trough plasma concentrations of bosentan increased 30-fold, resulting in severe headache, nausea, and vomiting, but no serious adverse events were reported. Mild decreases in blood pressure and increases in heart rate were reported.

Massive overdose may result in pronounced hypotension requiring active cardiovascular support. A case of overdose by an adolescent male patient with 10,000 mg of bosentan has been reported. He had symptoms of nausea, vomiting, hypotension, dizziness, sweating and blurred vision. He

recovered completely within 24 hours with blood pressure support. Note: bosentan is not removed through dialysis.

STORAGE

Store below 30°C

SUPPLY

Blister packs (triplex) of 1 X 15's/ box

PRODUCT REGISTRATION HOLDER

RANBAXY (MALAYSIA) SDN BHD

(A Sun Pharma Company)

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MANUFACTURER

Manufactured in India By:

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